

# Urban Wage Premium\*

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## **Abstract**

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\*People to thank...

# 1. Introduction

As of 2015, workers in urban areas earn a 16 percent premium over their non-urban counterparts. The urban wage premium has become an empirical fact, widely documented in the urban economics literature (see for example Rosenthal and Strange (2004) for a review). The literature has closely explored the role of high productivity individuals sorting into cities (Combes et al., 2008). For example Yankow (2006) documents that up to two-thirds of the urban wage premium is driven by differences in worker skills. The literature has also closely studied the role of differential returns to experience in urban areas as a source of the wage premium (Baum-Snow and Pavan, 2012). Outside of human capital explanations, researchers have explored the role of agglomeration economies and firm survival to explain the remaining gap (Glaeser and Maré (2001), Wheaton and Lewis (2002), Combes et al. (2012), Behrens et al. (2014)).<sup>1</sup> Agglomeration economies are an attractive explanation, yet determining the underlying mechanism that contributes to agglomeration is difficult. Indeed a large literature has sought to explain the causes and mechanisms of agglomeration.

According to Marshall et al. (1920) physical proximity and connections to firms are a drivers of agglomeration. Similarly, Duranton and Puga (2004) and Head and Mayer (2004) stress how trade costs may shape agglomeration and industrial location, and thus may partially explain the observed urban wage premium. In this paper, we directly explore how improvements in transportation that shape connections between locations, measured by market access, impact the estimated urban wage premium in the United States between 1940 and 2015. Empirically we begin by estimating regres-

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<sup>1</sup>Recently, work has also shown that urban industry-level concentration has led to a markdown of wages as firms exercise monopsony power Azar et al. (2020), thus, in the absence of concentration, the urban wage premium may be even larger than previously thought.

sions of individual wages on a set of individual characteristics and location and time fixed effects. This set of regressions enables us to partial out the portion of the wage gap due to underlying differences in the skill-education-experience composition of workers between urban and non-urban areas, while at the same time, netting out common firm survival trends that contribute to the wage gap. Similar to prior work, we find that individual characteristics account for approximately half of the urban wage premium and become increasingly important after 1980. We then construct a measure of market access following Donaldson and Hornbeck (2016) and Jaworski and Kitchens (2019) at the MSA level drawing on newly digitized atlases of the US Highway Network each decade 1940-2010. Our measure of market access allows us to consider how changes in a city's connectedness to other locations affects the wage premium. In general, we find that market access (MA) has a direct, positive impacts on wages, but the market access variable is uncorrelated with the urban variable, thus it has little to no impact on the urban wage premium. Importantly, we find that a portion of market access is collinear with urbanization, thus it is important to measure market access, net of the own market effect. This should be clear as by definition market access captures local productivity, population density, and amenities. Combined, the MA estimates highlight the importance of localized productivity differences, that we explore in more detail.

In particular, we are interested in understanding how changes in MA, induced by expansions of the highway system have influenced each MSA's position in the wage premium distribution. That is, as the highways have expanded, has it altered the rank of specific cities by the size of the premium. This exercise relates to work by Yankow (2006), who explores heterogeneity in the wage premium among large and small MSAs.

Finally, we seek to understand to what extent spatial sorting on unobservables, such as unobserved amenities or unobserved productivity, drives

the urban wage premium. To address concerns regarding selection on unobservables, we follow Altonji and Mansfield (2018) and control for MSA level averages of individual characteristics in additional specifications. The inclusion of these MSA-level averages should, in theory, net out the influence of unobservable productivity, for example, due to higher returns to experience in urban areas, as described by Baum-Snow and Pavan (2012). For example, consider the unobservable trait of working longer hours, as discussed in Weinberg et al. (2004). If workers in a particular city work longer hours due to an unobserved and time varying preference for longer work days, then this may be reflected in the urban wage premium. Controlling for the MSA-level averages nets out this type of sorting, however it may overcontrol by limiting the scope for other mechanisms. In this case, controlling for observables to address unobserved preferences for longer hours may limit the scope of referrals from neighbors, as discussed in Bayer et al. (2008). In addition to controlling for MSA-level averages of individual characteristics, we also estimate specifications where we control for MSA level housing values. We add housing value controls to limit the impact that unobserved amenities or land use restrictions have on wages.

When we add MSA-level averages of individual characteristics and MSA average housing values to our specifications, we find that the urban wage premium rises/falls by XYZ. Overall, our findings highlight the role that selection, both by individuals and firms, contributes to the urban wage premium. Indeed, after controlling for many sources of selection, we find that the unexplained wage premium is XX percent of the observed premium.

In what follows, we provide a brief background of why cities formed and how transportation has evolved over time. We then describe our data and empirical strategy.

## 2. Historical Background

### 2.1 The Rise of Cities

As of 2019 over 82 percent of Americans live in urban areas, up from just under 40 percent in 1900. Many factors explain the rise and growth of cities throughout the 20th Century. First, both biological and mechanical innovations in agriculture altered patterns of crop specialization and the optimal mix of labor and capital, freeing both land and labor for alternate uses (Caselli and Coleman II (2001), Michaels et al. (2012), Olmstead and Rhode (2001), Griliches (1963), and Jung (2020)). Second, negative health externalities due to crowding in cities has been largely mitigated through public health efforts. Early in US history, cities were dirty, unhealthy places that required a wage premium to attract workers to face risks associated with infectious diseases such as cholera, typhoid, yellow fever, and smallpox (Cutler and Miller (2005), Beach et al. (2016), Ferrie and Troesken (2008), Alsan and Goldin (2019), and Anderson et al. (2020)). As urban health improved, the need to attract workers with high wages subsided, however, as population density increased, so too did rents. Finally, there have been drastic changes in transportation that have facilitated the growth of cities.

Cities initially formed in places where it was convenient to trade and process agricultural goods due to natural advantages (Boustan et al., 2013). With improvements in technology, the scale and scope of cities has changed. For example, new cities sprang up near navigable rivers and canals to support the shipment and transshipment of goods. Later, the railroads bypassed many previous cities and led to new urban centers.<sup>2</sup> In the post-WWII era, there have been significant improvements in car and truck transportation that have further expanded and reshaped cities. For example, Rephann and

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<sup>2</sup>As Bleakley and Lin (2012) demonstrate, agglomeration forces allow cities to exist after their technological obsolescence.

Isserman (1994) and Duranton and Turner (2012) highlight the role of the interstate highways in expanding growth rates and city-level employment in the United States. In the context of shipping transportation improvements, Brooks et al. (2018) and Ducruet et al. (2019) highlight how population has expanded as a result of containerization, typically in land abundant cities. Outside of the United States, work by Faber (2014) and Baum-Snow et al. (2020) document that trunk highway systems in China shifted activity from rural to urban areas. While the existing literature has shown that cities grow as a result of better transportation, it is less clear whether the growth is accompanied by an increase in the urban wage premium, due for instance to new agglomeration economies.

In our empirical exercise below, we will test whether improvements in market access stemming from internal highway improvements contributes to both wage growth and the urban wage premium.

## 2.2 Highway Background

In 1940, there were 3.28 million miles of road in the United States.<sup>3</sup> Our data suggests that there were just over 358 thousand miles of federal and state aid highways in 1940. Of these, 126 thousand miles of US Highway were paved, 8.8 thousand miles were improved, but not paved, and over 2 thousand miles were dirt roads.<sup>4</sup> The remaining 220 thousand miles of road in 1940 were state highways, of which just over 60 percent was paved.

Since 1940, federal and state highway networks have improved dramatically. First, over 1 million new miles of road were added to the network, although much of this mileage is maintained by local governments and is

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<sup>3</sup><https://www.fhwa.dot.gov/policyinformation/statistics/2018/vmt421c.cfm>

<sup>4</sup>Our measure of mileage may differ from other measures given our data procedure used to develop time harmonized highway networks. For example, rather than deleting segments and recoding segments over time, which generates 126,000 miles of paved US Highway, tracing 1940 paved US Highways results in 127 thousand miles.

for local travel.<sup>5</sup> Of the 1 million new miles, approximately 50 thousand were built or upgraded to create the Interstate Highway System (IHS), which drastically increased travel speeds between places in the network. As highlighted in Michaels (2008), the IHS construction was especially active in the 1960s and 1970s. In addition to the construction of the IHS, other upgrades to state and non-IHS Federal-Aid Highways occurred.<sup>6</sup> These smaller improvements include straightening road alignments, upgrading road surfaces, road widening, and urban bypasses; all of which increase travel speeds. In the post WWII era, there has also been an effort to use road construction as a local development tool. Today there are over 220,000 miles in the National Highway System, reflecting expansions and improvements of both the IHS and the US Highway network.<sup>7</sup>

The improvements to the highway network have had meaningful impacts on travel times in the United States. In Figure 1, we highlight how the average travel time between all county pairs (4.6 million unique routes) in the United States has changed between 1940 and 2010.

In 1940, the average travel time was 27.67 hours. By 2010, the average travel time had fallen by 9.44 hours, to 18.23 hours, a 34 percent relative decline.<sup>8</sup> The largest drops in travel time coincide with the completion of major portions of the IHS occurring between 1960 and 1970. As we document below, the changes in travel time varied significantly across space. In part, this is due to the funding rules of the IHS, which disproportionately favored the South and West.

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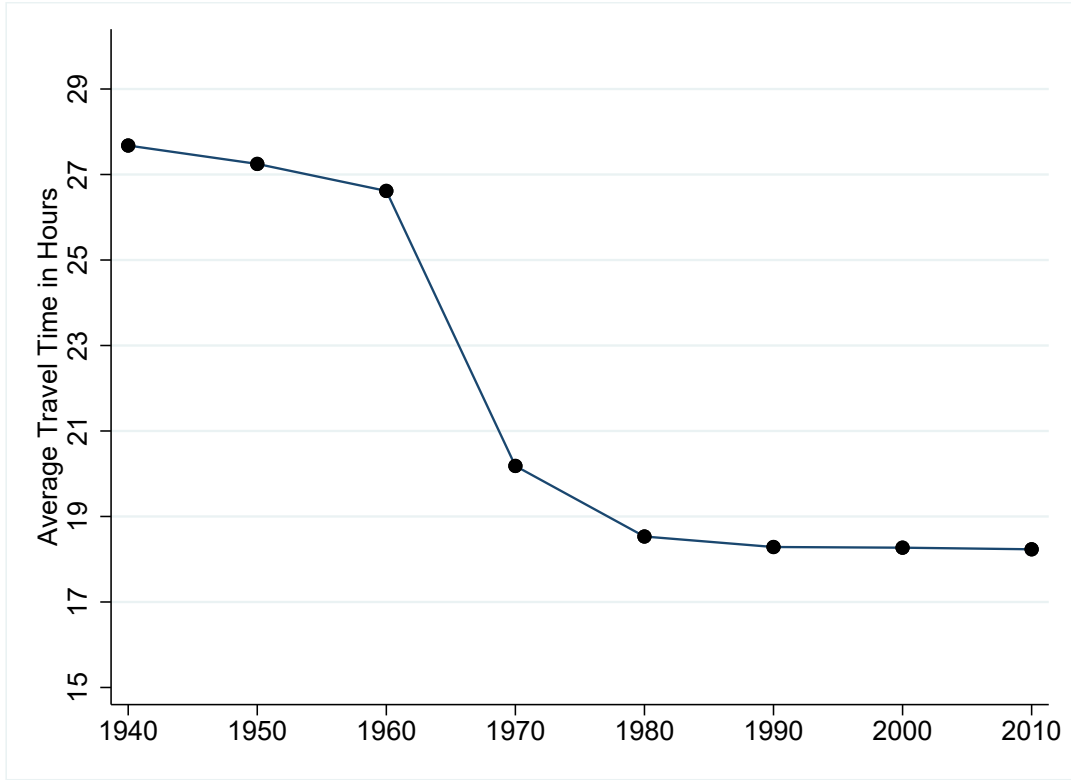
<sup>5</sup><https://www.fhwa.dot.gov/policyinformation/statistics/2018/vmt421c.cfm>

<sup>6</sup>Examples of non IHS improvements include the construction of the Appalachian Development Highway System, improvements to the Strategic Highway Network (STRAHNET), the Florida Turnpike, Texas Farm to Market (FM) Roads, among many others

<sup>7</sup><https://www.fhwa.dot.gov/policyinformation/statistics/2018/hm14.cfm>

<sup>8</sup>While travel times fell, the average route increased by 13 miles, or just over 1 percent.

Figure 1: Average Travel Time in Hours 1940-2010



*Notes:*

### 3. Data

To estimate the relationship between the urban wage premium and the role that market access has on the premium throughout the post war era, we draw on several datasets. First, we draw on publicly available microdata of decennial Censuses and the American Community Survey (ACS). We combine this individual level data with measures of market access constructed from a recently harmonized decadal dataset of the domestic US Highway Network that we describe in more detail below.

### 3.1 Wage and Individual Characteristic Data

To measure the average earnings per worker (wages), we use the IPUMS Full Count Census for 1940, the 1 percent sample Census for 1950 and 1970, the 5 percent sample Census for 1960, 1980, 1990, and 2000, and the 2005, 2010, and 2015 American Consumer Surveys. Our sample consists of adult men living in the continental US between the ages of 25 and 65 who are employed for at least 26 weeks in the sample year. We apply IPUMS person weights in our summary statistics and all our regressions.

We compute average weekly earnings (wages) by finding total income divided by the number of weeks worked in the given survey year. For individuals earning the top income code, we set their total income to the lower top income cutoff multiplied by 1.5.<sup>9</sup> If the urban wage premium is primarily driven by people in the top income group, our estimates of the urban wage premium will be biased downwards by this choice. All incomes are then multiplied by the Consumer Price Index to be in 2015 dollars. In order to have a consistent measure of occupation, we use the ‘occ1950’ variable provided by IPUMS which harmonizes occupation codes across all our survey years. In Table 1 we report the number of individual observations in each cross section.

Table 1 also reports the average weekly urban and nonurban wage in each time period. Two striking trends appear from the summary statistics. First, as we noted previously, the share of the population living in an urban areas has steadily increased between 1940 and 2015. This trend matches those highlighted in Boustan et al. (2013). Secondly, it is clear that both urban and nonurban mean weekly wages are growing over time in 2015 \$ from \$33.24 and \$25.23 in 1940 to \$1353.95 and \$1067.48 in 2015. The urban wage pre-

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<sup>9</sup>These cutoffs are reported in IPUMS INCWAGE documentation. In 2015 \$, the cutoffs are \$84,500 for 1940, \$99,200 for 1950, \$199,500 for 1960, \$309,000 for 1970, and \$225,000 for 1980. Afterwards, IPUMS reports income as the median wage above that year’s cutoff.

mium, the ratio of the urban to nonurban wage, follows a W shape. The urban premium starts at 0.32, and then falls sharply between 1940 and 1950, where it declines to a low of 19 percent in 1980. Since 1980, the urban wage premium has increased; in 2015 the premium was 26 percent.

Table 1: Summary Statistics

| Year | N          | % Urban | log(Market Access) |      | Urban Weekly Wage (2015 \$) |       | Nonurban Weekly Wage (2015 \$) |       |
|------|------------|---------|--------------------|------|-----------------------------|-------|--------------------------------|-------|
|      |            |         |                    |      |                             |       |                                |       |
|      |            |         | Mean               | SD   | Mean                        | SD    | Mean                           | SD    |
| 1940 | 16,939,585 | 0.66    | 3.39               | 0.87 | 33                          | 20    | 25                             | 17    |
| 1950 | 68,457     | 0.69    | 3.38               | 0.67 | 72                          | 41    | 59                             | 33    |
| 1960 | 1,364,970  | 0.66    | 3.33               | 0.66 | 124                         | 77    | 100                            | 60    |
| 1970 | 280,338    | 0.75    | 3.42               | 0.70 | 205                         | 138   | 166                            | 105   |
| 1980 | 1,910,440  | 0.73    | 3.22               | 0.57 | 409                         | 284   | 341                            | 221   |
| 1990 | 2,262,149  | 0.74    | 3.12               | 0.46 | 701                         | 562   | 558                            | 405   |
| 2000 | 2,583,088  | 0.79    | 3.08               | 0.42 | 1,008                       | 1,015 | 767                            | 677   |
| 2005 | 530,737    | 0.79    | 3.08               | 0.42 | 1,143                       | 1,146 | 863                            | 712   |
| 2010 | 530,783    | 0.80    | 3.01               | 0.51 | 1,228                       | 1,210 | 961                            | 819   |
| 2015 | 561,934    | 0.81    | 3.00               | 0.52 | 1,353                       | 1,446 | 1,068                          | 1,011 |

*Notes:* Summary statistics weights individuals by their person weight. Weekly wage is reported in 2015 \$. Market access is calculated from solving  $MA_c = \rho \sum_d \tau_{cd}^{-\theta} MA_d^{-1} Y_d$ , where transportation costs,  $\tau_{cd}$ , are calculated as described by Equation ??.

## 4. The Urban Wage Premium: 1940-2010

### 4.1 Regression Framework

Our goal in this paper is two-fold. First, we aim to describe the observed urban wage premium from a historical perspective. For this purpose, we

estimate the urban wage premium from 1940 to 2010 by comparing average weekly wages for people who live in urban and non-urban areas.<sup>10</sup> For each census survey year, we run the following basic regression:

$$\log(\text{Wage}_i) = \beta_0 + \beta_1 \text{Urban}_i + \varepsilon_i, \quad (1)$$

where  $\text{Urban}_i$  is an indicator if worker  $i$  resides in an urban area (as classified by the Census Bureau). Therefore, our estimate of  $\beta_1$  captures the percent difference in weekly wages earned by urban-residing workers relative to non-urban-residing workers. These estimates will serve as a baseline description of the differences in wages that we *observe* in our survey data and are not interpreted as the “causal effect” of living in an urban area.

One potential difficulty in comparing estimates across years is that the Census Bureau classifies new (typically smaller) urban areas throughout the decades. There is evidence that the urban wage premium grows with city size and therefore we might expect to see a declining premium over time due to an expansion of urban area status.<sup>11</sup> In a robustness exercise in the Online Appendix, Figure B.1 reestimates our specification using only urban areas as classified in 1940. The results are very similar in this specification.

The second goal of this paper is to understand the different sources underlying the wage premium. The unadjusted urban premium reflects a combination of a true causal wage premium; differences in skills due to endogenous selection of labor into urban areas; differential returns to labor market experience; and other potential city labor amenities such as agglomeration economies. To better understand how these different channels impact the urban wage premium, we extend our regression specification to control for confounding variables. Specifically, we estimate a set of regressions for each

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<sup>10</sup>Weekly wages are adjusted for inflation using the BLS’ consumer price index in each survey year.

<sup>11</sup>For example, see Glaeser and Maré (2001).

census survey year as follows:

$$\log(\text{Wage}_i) = \beta_0 + \beta_1 \text{Urban}_i + X_i \delta + \varepsilon_i, \quad (2)$$

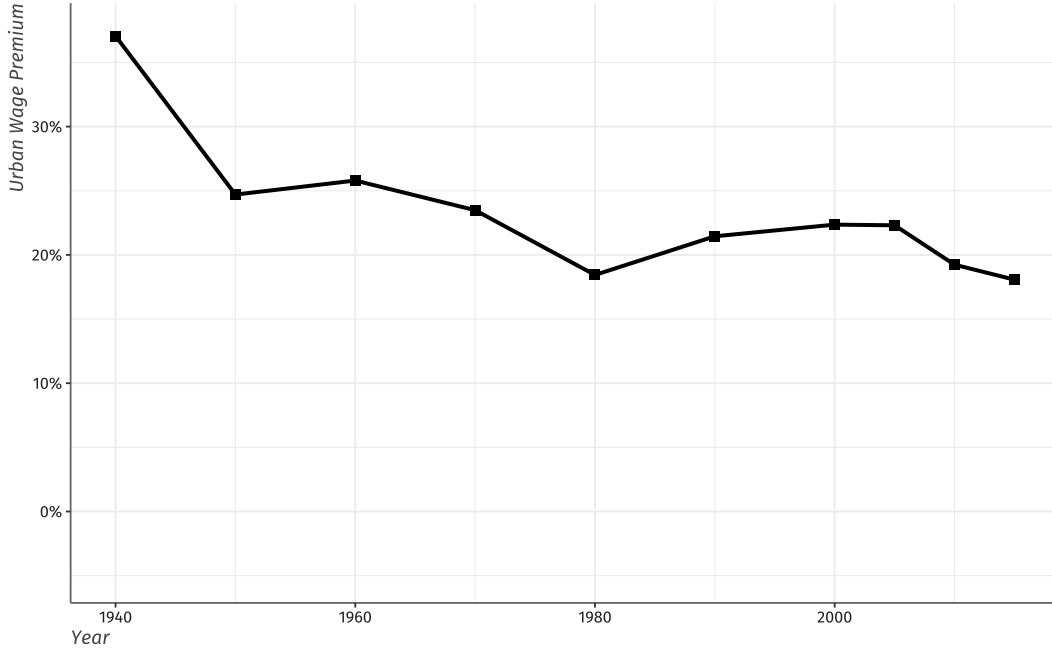
where  $X_i$  is a vector of control variables for worker  $i$ . To facilitate consistent comparisons across years, we use the same set of individual-level covariates in each survey-year. This limits the number of variables we include to a set of 5-year age bins, indicator variables for education level, and an indicator for being in a racial minority. Therefore our set of control variables are not able to control for all potential sources of endogenous sorting. For example, *unobserved* individual attributes like motivation and aptitude can lead to sorting but not be controlled for by the econometrician (Yankow (2006) and Combes et al. (2008)). Thus, we do not provide a causal interpretation to the estimate of  $\beta_1$  in equation (2) either. The aim of this exercise is to determine the amount of the overall observed wage premium that these observable covariates are able to explain and how that may change over time.

In section 5, we aim to provide a causal estimate of the urban wage premium over time by controlling for *all* sources of endogenous sorting into cities. Since we do not observe the same worker in multiple survey years, we cannot follow previous papers in including individual-specific fixed effects. Instead, we follow the methodology recently proposed by Altonji and Mansfield (2018) which provides conditions under which controlling for group-averages of observed variables controls for *all* the unobserved covariates that we don't have individual-level data for.

## 4.2 Results

The results of estimating equation (1) is displayed in Figure 2. The urban wage premium starts around 35% in 1940, decreases to 25% between 1950 and 1970, and starting in 1980 the premium converges to where it stands

Figure 2: Unadjusted Urban Wage Premium 1940-2015



Notes: Each point is the log of the ratio of average wage among urban male wage-earners and non-urban male wage-earners in the given survey year.

today at around 20%. Our results match the estimates found in Figure 3 of Boustan et al. (2013) for estimates before 1980 while our estimates in later years are smaller than their estimates.<sup>12</sup> Our estimates are close to others in the literature. Our unadjusted difference for the year 1990 and 2000 is smaller in magnitude than the estimate of between 21 and 32% from Glaeser and Maré (2001) and similar in magnitude to 1980 and 2000 as documented

<sup>12</sup>The reason for the difference in estimates comes from our use of a regression framework. The method in Boustan et al. (2013) estimates the following

$$\log(\hat{\mathbb{E}}[\text{Wage}|\text{Urban} = 1]) - \log(\hat{\mathbb{E}}[\text{Wage}|\text{Urban} = 0])$$

which differs from the regression estimate

$$\hat{\beta}_1 = \hat{\mathbb{E}}[\log(\text{Wage})|\text{Urban} = 1] - \hat{\mathbb{E}}[\log(\text{Wage})|\text{Urban} = 0].$$

. The latter is a more precise estimate for average percent difference in wages.

in Baum-Snow and Pavan (2012) and Boustan et al. (2013). Overall, our unadjusted estimates at the urban wage premium provide evidence for the overall decline in the ‘importance’ of cities for wages over time, but do not provide evidence of a ‘modern resurgence’ of the wage premium that is suggested by previous descriptive and historical work.

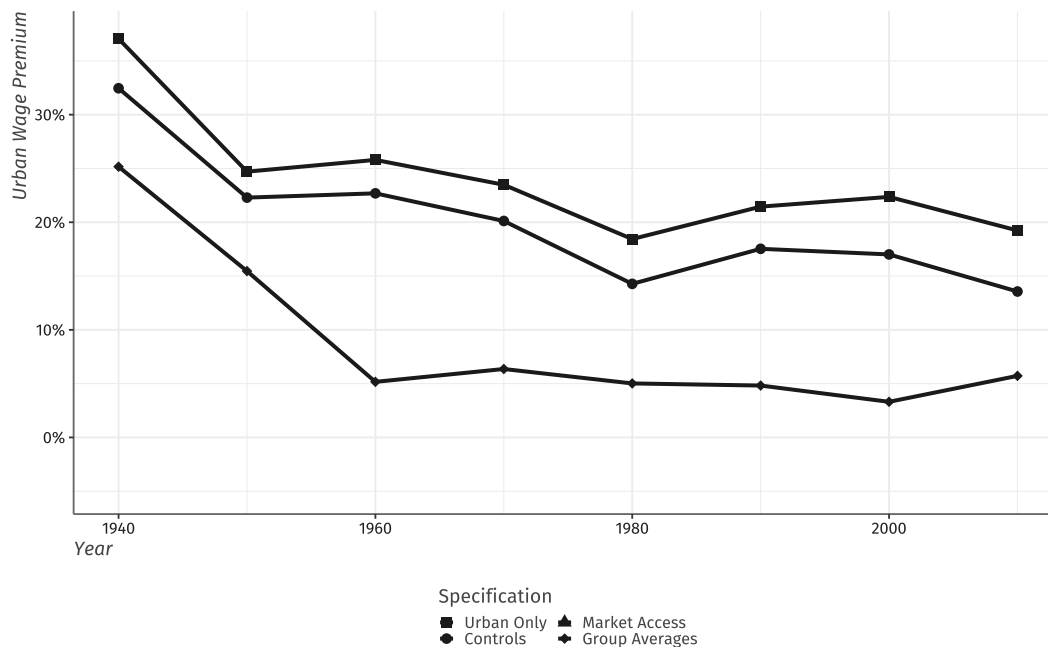
We report the urban wage premium estimates after controlling for individual characteristics in Figure 3 by the line with circle markers along with the original estimates marked with squares. Our estimates suggest that in 1940, the individual-level controls explain about 1/8 of the unadjusted urban wage premium, but the shape of the premium over time remains consistent. However, the explanatory power of individual controls steadily grows over time. In 2015, the individual-level controls explain about 1/3 of the urban premium highlighting the importance of sorting of individuals in explaining wage premiums in the modern era. Across all years in the sample, we find that individual-level observables explain between 8% and 33% of the urban wage premium, with the individual characteristics explaining a larger portion of the premium since 1980.

## **5. The Causal Effect of Urban Areas**

### **5.1 Methodology**

Although we include a small set of individual-level control variables, there are many sources of unobserved covariates that can sort between urban and non-urban areas and lead to differences in average pay. We use the method proposed by Altonji and Mansfield (2018) to control for these unobserved variables. The proposed method relies on two key relationships in how individuals sort. First, group-level averages of covariates in urban and non-urban areas reveal information about the differences in various amenities

Figure 3: Urban Wage Premium 1940-2015



*Notes:* This figure plots the coefficient estimates for survey-year level estimates of the urban wage premium. In the grey line with square markers, the estimate is found by taking the ratio of the average log weekly wage for urban and nonurban workers. The remaining lines run versions of the regression given by Equation 2. In the line with diamond markers, there are no controls and no measure of market access; the line with triangle markers includes controls for age in 5-year bins, education levels, and an indicator for being White; the line with circle markers includes the same controls and the log of market access.

that cause people to sort between urban and non-urban areas.<sup>13</sup> That is, the averages of covariates in urban and non-urban areas are correlated with average amenity quantities in urban and non-urban areas. For example, Diamond and Moretti (2021) find that higher-educated workers prefer a lot of amenities that are more abundant in cities like more food options. Second, individuals sort based on their unobserved covariates because of the *same* set of amenities. That is, the average of unobserved covariates are also cor-

<sup>13</sup>This can even include the wage premium itself.

related with the average amenity quantities. Continuing with our previous example, preferences for more food options might be correlated with unobservable qualities like curiosity that can lead to higher productivity in the form of innovation. Hence, controlling for the average of observable covariates (e.g. education) should therefore, at least partially, control for averages of unobservable covariates (e.g. curiosity).

Altonji and Mansfield (2018) show that under a set of conditions discussed below (and more formally in the authors' work), there exists a *linear* mapping between group-level averages of *observed* covariates and group-level averages of *unobserved* covariates.<sup>14</sup> Therefore including group-level averages linearly in Equation 2 will absorb *all of* the observed urban wage premium that is due to sorting on unobservable covariates.<sup>15</sup>

The result proposed by Altonji and Mansfield (2018) relies on a few assumptions. First, it is assumed that individuals choose locations following a logistic choice model with a latent function that models the appeal of locations as a function of both the observed and unobserved covariates. This is standard in the urban economics literature. The second assumption, called the 'rank condition' in their paper, pertains with the set of factors that drive sorting along the observable group averages. Essentially, the set of factors that drive sorting on *observable* covariates must contain *all* factors that drive sorting based on *unobservable* covariates. The intuition for this is that group averages reveal information on the amenity factors that cause sorting based on observed covariates. If a different set of amenity factors drive sorting based on the unobservable characteristics, then information on these group averages do not inform us on the unobservable characteristics.

The second part of the rank condition is that the number of group-average

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<sup>14</sup>This is true even when there is no correlation between unobservable and observable variables at an individual-level.

<sup>15</sup>The group-level averages also will absorb the effects of the amenities that drive sorting, so changes in the wage premium coefficient reflect removal of both sources.

variables must be larger than the total number of factors driving sorting into urban areas. The intuition for this is that regression will first ‘invert’ the observed group-level averages to amenities, but this inversion is only possible if the mapping is one-to-one. If there were more amenities than group-average variables, then there would not be a unique inverse and therefore the mapping between observable averages and unobservable averages would break. This second condition warrants more discussion as initially it may seem that there are many amenities that drive sorting into or out of urban areas. However, these amenities are often highly correlated with one another. More formally, the rank condition requires the effective rank of the group-average variables be larger than the effective rank of amenity variables. Diamond (2016) provide evidence using a principal component analysis on a very detailed set of amenities that, at least in the 21st century, the effective rank of amenities is around 5 (with three components explaining most of the variation). Therefore, one of our identifying assumption on our set of group-average variables to have a large enough effective rank to satisfy the rank condition which is estimated around 3-5. If this were not the case, then we would only *partially* absorb the effects of individual sorting.

The group-level averages we include are the share of individuals of different racial/ethnic groups, the share of individuals at different levels of educational attainments, the share of individuals who were veterans, the share of individuals of different marital status, and a measure of market access which summarizes a location ability to trade with other areas.

## 5.2 Results

The results of this exercise are presented in Figure 3 by the line with diamond markers. First, we turn to estimates after 1990, as we can directly compare these to similar estimates found in the literature. If our identifying assump-

tions hold true, then our estimates reflect the urban wage premium after controlling *completely* for individual characteristics. We find an urban wage premium between 1990 and 2010 between -2% and 5%. These results fall very close to estimates in the literature that use panel data with individual fixed-effects to control for individual sorting. Glaeser and Maré (2001) estimate a premium of around 5% in the 90s; Yankow (2006) estimate a premium between 3-5% and D'Costa and Overman (2014) estimate a premium between 2-5% in the 2000s. Our estimates falling so close to these estimates provide support for our empirical approach. Overall, our results find that after 1990, there is very little causal benefits to living in the city.

Now, we extend this literature by analyzing the premium from a historical perspective. The estimates of around 5% continue back until 1960. However, in 1940 the premium remains around 25% and around 15% in 1950. This evidence suggests that historically cities indeed did provide large benefits to any individuals who came there to work. This is consistent with historical accounts that ...

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## **A. Full Urban Wage Premium Regressions**

Full results of our regression estimates in Figure 3 can be found in Table A.1.

Table A.1: All Urban Wage Premium Estimates by Survey Year

|                                                                           | 1940               | 1950               | 1960               | 1970               | 1980               | 1990               | 2000               | 2005               | 2010               | 2015               |
|---------------------------------------------------------------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|
| <b>Panel A: Urban only</b>                                                |                    |                    |                    |                    |                    |                    |                    |                    |                    |                    |
| Urban                                                                     | 0.3153<br>(0.0177) | 0.2208<br>(0.0131) | 0.2295<br>(0.0135) | 0.2109<br>(0.0119) | 0.1692<br>(0.0110) | 0.1943<br>(0.0157) | 0.2018<br>(0.0144) | 0.2014<br>(0.0160) | 0.1759<br>(0.0165) | 0.1661<br>(0.0175) |
| N                                                                         | 16,939,585         | 68,457             | 1,364,970          | 280,338            | 1,910,440          | 2,262,149          | 2,583,088          | 530,737            | 530,783            | 561,934            |
| <b>Panel B: Urban with Controls</b>                                       |                    |                    |                    |                    |                    |                    |                    |                    |                    |                    |
| Urban                                                                     | 0.2808<br>(0.0170) | 0.2012<br>(0.0125) | 0.2045<br>(0.0123) | 0.1833<br>(0.0119) | 0.1334<br>(0.0116) | 0.1615<br>(0.0162) | 0.1571<br>(0.0125) | 0.1517<br>(0.0131) | 0.1272<br>(0.0135) | 0.1108<br>(0.0137) |
| N                                                                         | 16,939,585         | 68,457             | 1,364,970          | 280,338            | 1,910,440          | 2,262,149          | 2,583,088          | 530,737            | 530,783            | 561,934            |
| <b>Panel C: Urban, Controls, and Market Access</b>                        |                    |                    |                    |                    |                    |                    |                    |                    |                    |                    |
| Urban                                                                     | 0.146<br>(0.014)   | 0.117<br>(0.014)   | 0.109<br>(0.012)   | 0.093<br>(0.011)   | 0.071<br>(0.014)   | 0.065<br>(0.016)   | 0.085<br>(0.014)   | 0.083<br>(0.015)   | 0.085<br>(0.015)   | 0.076<br>(0.018)   |
| $\log(MA)$                                                                | 0.093<br>(0.009)   | 0.079<br>(0.016)   | 0.104<br>(0.020)   | 0.089<br>(0.012)   | 0.078<br>(0.021)   | 0.186<br>(0.026)   | 0.152<br>(0.024)   | 0.147<br>(0.028)   | 0.089<br>(0.024)   | 0.074<br>(0.025)   |
| N                                                                         | 16,939,585         | 68,457             | 1,364,970          | 280,338            | 1,910,440          | 2,262,149          | 2,583,088          | 530,737            | 530,783            | 561,934            |
| <b>Panel D: Urban, Controls, and Market Access (excluding own market)</b> |                    |                    |                    |                    |                    |                    |                    |                    |                    |                    |
| Urban                                                                     | 0.254<br>(0.012)   | 0.190<br>(0.014)   | 0.194<br>(0.013)   | 0.186<br>(0.012)   | 0.134<br>(0.011)   | 0.151<br>(0.014)   | 0.147<br>(0.011)   | 0.145<br>(0.011)   | 0.122<br>(0.012)   | 0.103<br>(0.012)   |
| Remove-own $\log(MA)$                                                     | -0.020<br>(0.010)  | 0.019<br>(0.007)   | -0.034<br>(0.019)  | 0.010<br>(0.017)   | 0.007<br>(0.014)   | 0.108<br>(0.037)   | 0.102<br>(0.028)   | 0.077<br>(0.032)   | 0.059<br>(0.034)   | 0.076<br>(0.035)   |
| N                                                                         | 16,939,585         | 68,457             | 1,364,970          | 280,338            | 1,910,440          | 2,262,149          | 2,583,088          | 530,737            | 530,783            | 561,934            |

*Notes.* Each cell is the coefficient estimate corresponding to the regression for the survey year indicated by the column label. The dependent variable is log weekly earning and each regression includes an indicator for living in an urban area. Panel A just includes the Urban indicator; Panel B includes controls for age in 5-year bins, education levels, and an indicator for being White; Panel C includes the above controls and our measure of market access; Panel D modifies the  $\log(MA)$  measure by removing each unit's market access in equation ?? in Appendix subsection ?? All regressions are weighted by the IPUMS person weights. Standard errors are in parenthesis and are clustered at the MSA-Nonurban State level

## **B. MSA-level Market Access**

We aggregate our county-level measure of Market Access from Equation ?? to a measure at the urban area. For each year, we start with a crosswalk provided by IPUMS that lists which counties fall (at least partially) inside the boundaries of the metro area. Our metro area boundaries are the 1950 Standard Metropolitan Area for 1940 and 1950, the Metropolitan Statistical Areas for their respective year in 1960-2010, and the Core-Based Statistical Area for 2015.

This list is modified slightly to include MSAs that are recorded in the IPUMS survey data but are not included in that year's crosswalk. The counties that the MSA covers are identified by hand using the shape files from NHGIS for that year or if missing from the previous year.

For a given year, a metro area's market access is the geometric average of the corresponding counties' market access. Then, for our measure of market access for the nonurban portion of a state is the geometric average of all counties that are not within any metro areas. We include robustness checks for weighting the average by county population.

### **B.1 Robustness Checks**

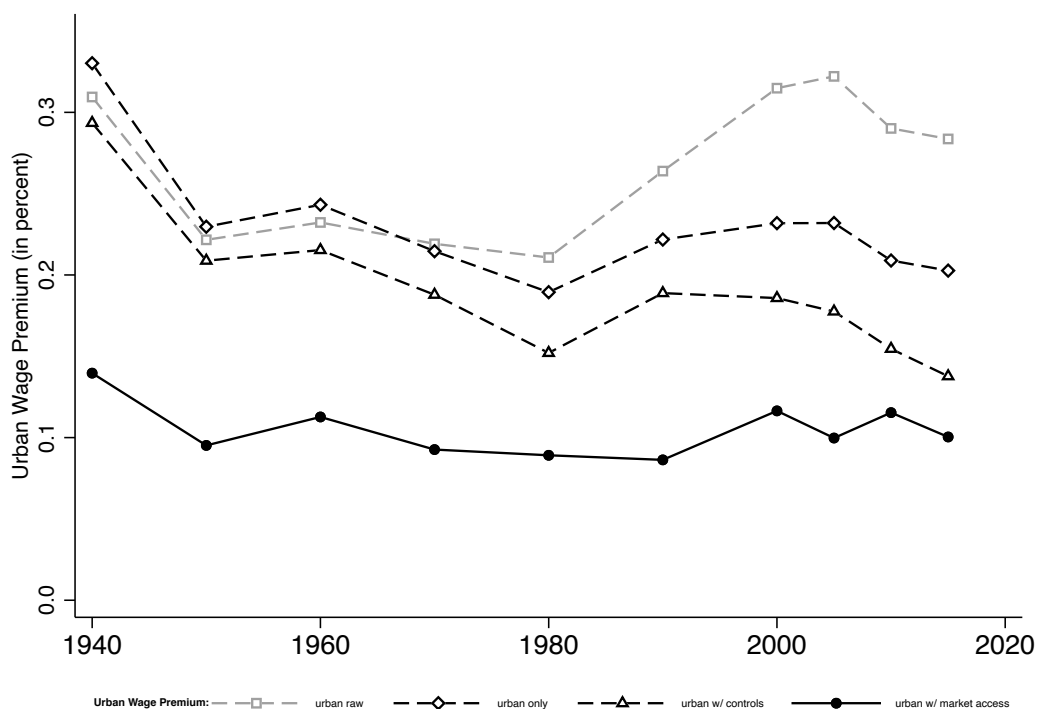
#### **B.1.1 MSA Selection Over Time**

Over decades as places grew in size, more and more places became classified as urban areas and these areas tended to be smaller than long-standing MSAs. There could be concern that the observed decline in the urban wage premium as smaller cities are added over time, the observed decline in the urban wage premium is driven by the addition of smaller cities over time. Figure B.1 shows this is not the case.

To create this, we keep all MSAs that exist in the entire sample from 1940 -

2015 (removing those that are added over time and the few that are lost over time). Observations are dropped for individuals identified in an MSA that does not exist for the entire sample. There still remains the issue that MSAs tend to grow over time incorporating less densely populated areas over time which we are unable to fix because the survey data does not feature exact locations of individuals.

Figure B.1: Urban Wage Premium 1940-2015 for 1940 MSAs



*Notes:* This figure plots the coefficient estimates for survey-year level estimates of the urban wage premium. In the grey line with square markers, the estimate is found by taking the ratio of the average log weekly wage for urban and nonurban workers. The remaining lines run versions of the regression given by Equation 2. In the line with diamond markers, there are no controls and no measure of market access; the line with triangle markers includes controls for age in 5-year bins, education levels, and an indicator for being White; the line with circle markers includes the same controls and the log of the weighted market access.